Practice workbook · Identify a measurement weakness and improve accuracy

Methods with subjective scoring, uneven bubble counts, too few or widely spaced intervals, or a contaminated reading.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-PRAC-EVALUATE-MEASUREMENT-ACCURACY

1. Question 1 · part a.iv · 3 marks

November 2021 Paper 1

Worked-example retry · Attempt part a.iv only.

1 The enzyme catalase is found in the liver.

Catalase acts as a catalyst when added to hydrogen peroxide solution, causing the solution to break down into water and oxygen.

A student investigates the increase in the rate of breakdown of hydrogen peroxide into water and oxygen using catalase.

This is the student's method.

- put 100 g of liver into 100 cm^3 of water, and mix into a smooth paste
- put 10 cm^3 of the paste into a beaker at room temperature
- add one drop of hydrogen peroxide to the paste
- assess how many oxygen bubbles are produced on a scale of 0 to 5, with 5 representing most bubbles and 0 representing no bubbles

The student repeats the investigation using different conditions.

The conditions he uses are:

- liver paste heated to 70 °C
- liver paste that has been kept at 3 °C for an hour
- acidified liver paste
- liver paste with an alkali added
- liver paste with no hydrogen peroxide added

These are the student's results.

room temperature = 5 heated to 70 °C = 0

kept at 3 °C = 1 acid added = 2 alkali added = 3

no hydrogen peroxide added = 0

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1. Question 1 · part a.iv · 3 marks

November 2021 Paper 1

Worked-example retry · Attempt part a.iv only.

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(a) (i) Give the student's results in the form of a suitable table.

(3)

(ii) State why this investigation is not reliable.

(1)

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(iii) Give a reason why the results of this investigation are not accurate.

(1)

(iv) Describe how the investigation could be improved so that the conclusions could be more accurate and reliable.

(3)

1. Question 1 · part a.iv · 3 marks

November 2021 Paper 1

Worked-example retry · Attempt part a.iv only.

(b) (i) State the condition where the enzyme was most active.

(1)

(ii) State the reason for including the beaker with no hydrogen peroxide added.

(1)

(iii) Explain the result obtained when the conditions were acidic.

(4)

(Total for Question 1 = 14 marks)

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1. Question 1 · part a.iv · 3 marks · Working and answer

Working and answer

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2. Question 1 · part a.iii · 1 mark

November 2021 Paper 1

Further practice · Attempt part a.iii only.

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The conditions he uses are:

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- liver paste with no hydrogen peroxide added

These are the student's results.

room temperature = 5 heated to 70 °C = 0

kept at 3 °C = 1 acid added = 2 alkali added = 3

no hydrogen peroxide added = 0

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2. Question 1 · part a.iii · 1 mark

November 2021 Paper 1

Further practice · Attempt part a.iii only.

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(3)

(ii) State why this investigation is not reliable.

(1)

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(iii) Give a reason why the results of this investigation are not accurate.

(1)

(iv) Describe how the investigation could be improved so that the conclusions could be more accurate and reliable.

(3)

2. Question 1 · part a.iii · 1 mark · Working and answer

Working and answer

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3. Question 7 · part b.iv · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.iv only.

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7 Chymosin is an enzyme found in young humans, but not in adults. It converts soluble proteins in milk into solid proteins.

(a) (i) Suggest the advantages to a young human of having chymosin in their alimentary canal.

(2)

(ii) Explain why chymosin is only needed in young humans.

(2)

(iii) Describe a test to show that the solid formed by the action of chymosin on milk is a protein.

(3)

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3. Question 7 · part b.iv · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.iv only.

(b) A student investigates the effect of carbon dioxide on the activity of chymosin.

The student bubbles different volumes of carbon dioxide gas into five samples of milk.

He then adds chymosin to each sample and records the time taken for the milk protein to become solid.

The student repeats this method three times.

The table shows the student's results.

Number of bubbles of CO_2	Time taken for chymosin to turn milk protein solid, in seconds				
	Test 1	Test 2	Test 3	Test 4	Mean
100	253	257	250	260	255
150	238	232	241	229	
200	216	214	219	211	215
250	208	202	212	198	205
300	210	200	199	311	203

(i) State three factors that the student should control.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(ii) Calculate the missing mean (average) time taken for 150 bubbles.

(2)

mean time taken = ...

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P 6 2 0 5 4 A 0 2 0 2 4

3. Question 7 · part b.iv · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.iv only.

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(iii) In test 4 there is an anomalous result.

State how the student deals with this result.

(1)

.....

.....

(iv) State why recording the number of bubbles may produce inaccurate results.

(1)

.....

.....

(Total for Question 7 = 14 marks)

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Working and answer

4. Question 7 · part b.ii · 2 marks

June 2019 Paper 1R

Further practice · Attempt part b.ii only.

7 Cheese is processed from milk. During this process, an enzyme is used to convert the lactose in milk into lactic acid.

A student uses this method to investigate the process.

- dissolve lactose in water to produce a solution
- put the lactose solution into three different test tubes
- keep the first test tube at 2 °C, the second test tube at 25 °C and the third test tube at 55 °C
- add enzyme solution to the lactose solution in each test tube
- measure the amount of lactic acid in the solution every 10 hours for a total of 60 hours

The table shows the student's results.

Time after adding enzyme in hours	Amount of lactic acid in the solution in arbitrary units		
	2 °C	25 °C	55 °C
0	0.1	0.1	0.1
10	0.1	0.2	0.2
20	0.1	0.3	0.2
30	0.2	0.8	0.4
40	0.3	1.8	0.7
50	0.4	2.2	1.0
60	0.6	2.3	1.3

(a) Describe what additional information another student would need to repeat this investigation.

(4)

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4. Question 7 · part b.ii · 2 marks

June 2019 Paper 1R

Further practice · Attempt part b.ii only.

(b) (i) Estimate the optimum temperature for this enzyme.

(1)

(ii) Explain why it is not possible to give an accurate value for the optimum temperature using the student's method.

(2)

(c) The student continued the investigation for a further 60 hours.

Explain the results that would be obtained.

(3)

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4. Question 7 · part b.ii · 2 marks · Working and answer

Working and answer

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5. Question 6 · part a.iii · 2 marks

January 2023 Paper 1

Further practice · Attempt part a.iii only.

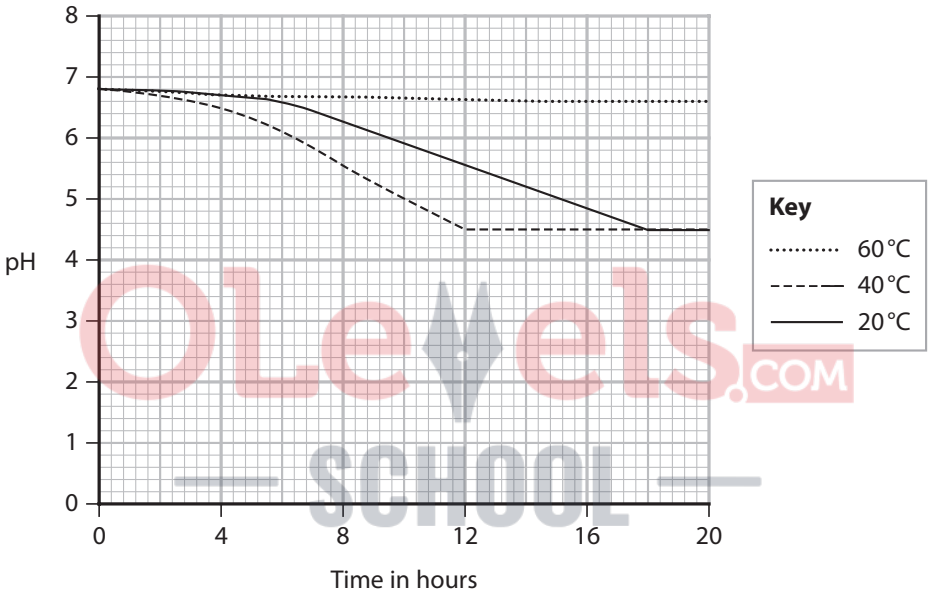
6 Yoghurt can be produced by incubating milk with a small amount of live yoghurt.

The live yoghurt contains *Lactobacillus* bacteria.

Enzymes convert lactose in the milk into lactic acid causing the pH to decrease to around pH 4.4.

In an investigation, the change in pH as yoghurt is produced was measured during 20 hours at three different temperatures.

The graph shows the results.



(a) (i) Calculate the difference in the time taken to produce yoghurt at 40 °C instead of 20 °C.

(2)

time = ... hours

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5. Question 6 · part a.iii · 2 marks

January 2023 Paper 1

Further practice · Attempt part a.iii only.

(ii) Give two variables that need to be controlled in this investigation.

(2)

1 .....

.....

2 .....

.....

(iii) Describe how this investigation could be improved to find the optimum temperature for the production of yoghurt.

(2)

.....

.....

.....

.....

.....

(iv) In some methods to produce yoghurt, the milk is boiled and then cooled before the bacteria are added.

Give the reason for boiling and the reason for cooling the milk.

(2)

boiling

.....

.....

cooling

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.....

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5. Question 6 · part a.iii · 2 marks · Working and answer

Working and answer

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6. Question 3 · part c.ii · 1 mark

June 2019 Paper 2

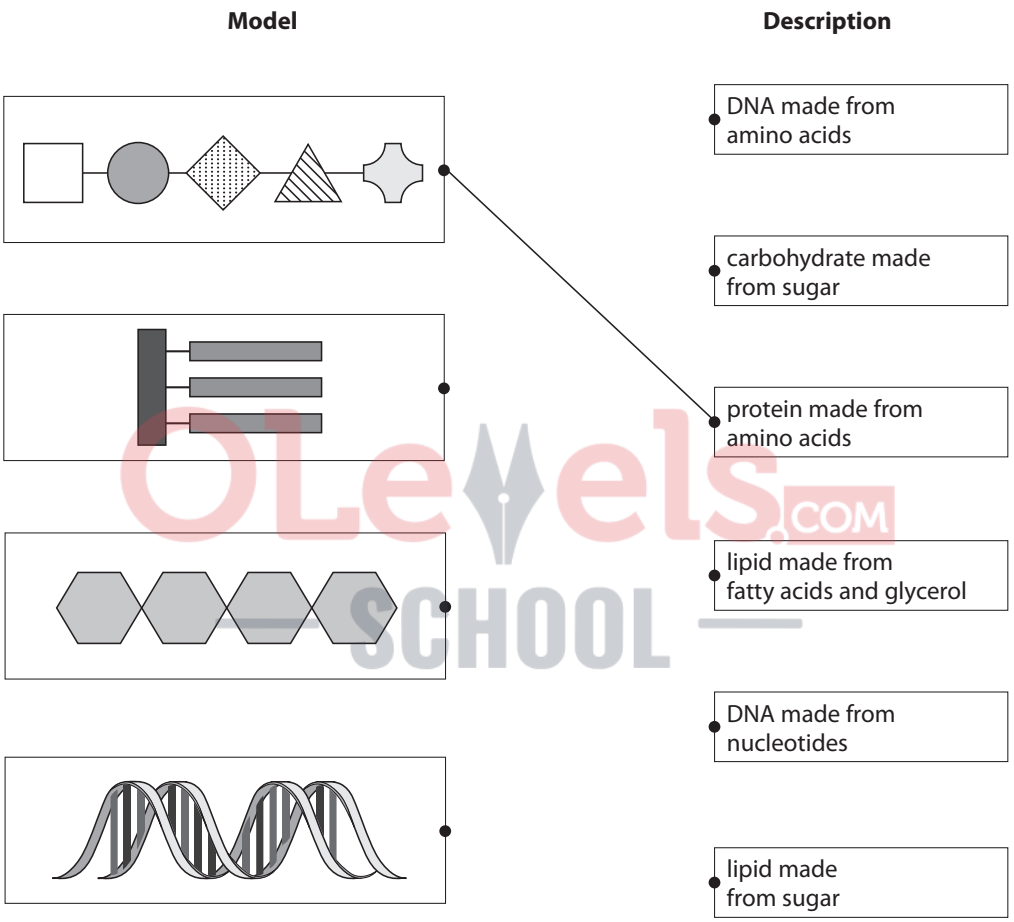
Further practice · Attempt part c.ii only.

3 (a) The diagram shows models of four biological molecules.

Draw a straight line from each model to its correct description.

One has been done for you.

(3)



(b) Which body organ produces enzymes that break down protein?

(1)

- ☒ **A** gall bladder
- ☒ **B** large intestine
- ☒ **C** mouth
- ☒ **D** stomach

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6. Question 3 · part c.ii · 1 mark

June 2019 Paper 2

Further practice · Attempt part c.ii only.

(c) A student uses this method to test different substances for protein.

- grind solid substances into small pieces
- place substances into separate test tubes
- add a few drops of Biuret reagent to each substance
- record the colour change for each substance

The table shows the student's results.

Substance	Colour after Biuret test
milk	purple
pasta	blue
lemon juice	blue
cheese	purple
distilled water	purple

(i) In the student's test, how many of the substances give a positive result for protein?

(1)

.....

(ii) The result for distilled water is incorrect.

Suggest one mistake the student could have made to get this incorrect result.

(1)

.....

.....

(d) The table shows an incomplete risk assessment for the Biuret test.

Complete the table by describing how to reduce the risk of each hazard.

(2)

Hazard	Reducing risk
stools – trip hazard	keep stools under bench
broken glass – cuts	
Biuret reagent – irritant	

(Total for Question 3 = 8 marks)

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6. Question 3 · part c.ii · 1 mark · Working and answer

Working and answer

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